

Lecture 7 - Routh-Hurwitz and Sys Type

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7.1 System Type

- A stable system is of Type k if it can track polynomial inputs of degree k with bounded error.
- System type let's us characterize tracking performance for polynomial signals.

Example to illustrate:

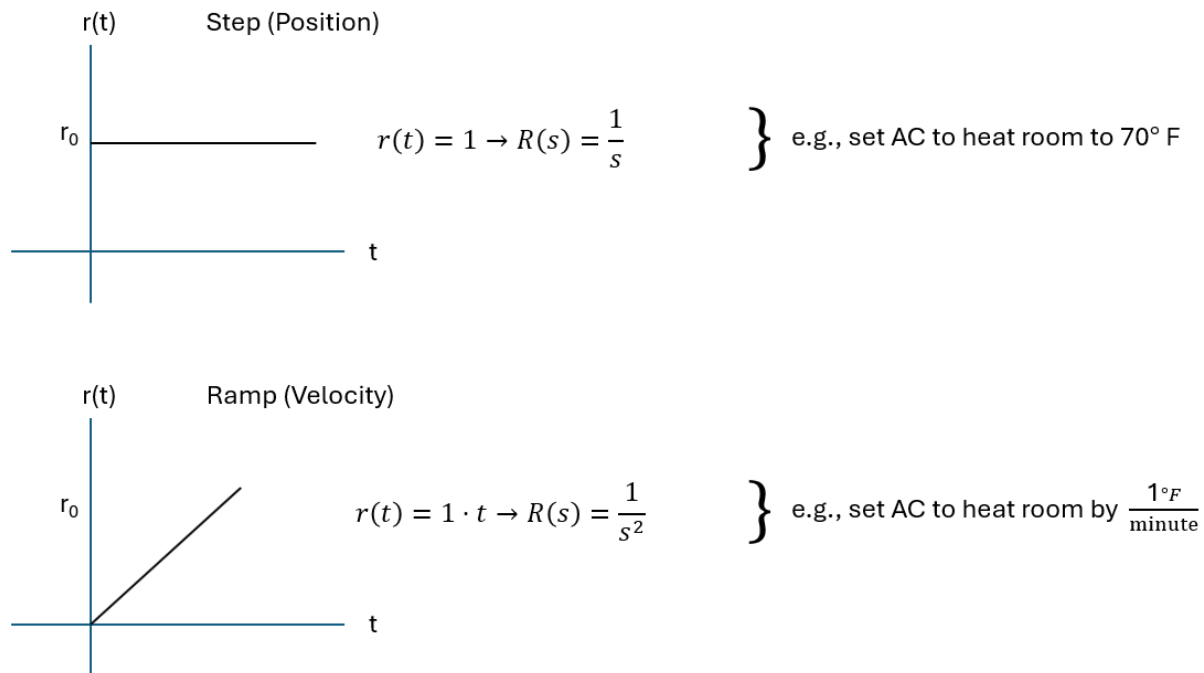


Figure 7.1: Example of system types.

Consider plant $G = \frac{1}{s+a}$, and let's use proportional feedback:

$$H(s) = k_p \rightarrow G_{RY} = \frac{GH}{1+GH} = \frac{\frac{k_p}{s+a}}{1 + \frac{k_p}{s+a}} = \frac{k_p}{s+a+k_p}$$

$$Y(s) = G_{RY}(s)R(s) \rightarrow E(s) = R(s) - Y(s) = (1 - G_{RY})R(s) = \underbrace{\frac{s+a}{s+a+k_p}}_{G_{RE}} R(s)$$

For Step input, FVT: (for tracking)

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = s \cdot \frac{1}{s} \cdot \frac{s+a}{s+a+k_p} = \frac{a}{a+k_p}$$

$$\therefore \text{We track with bounded error: } E_{ss} = \frac{a}{a+k_p}$$

For Ramp input, FVT:

$$\lim_{s \rightarrow 0} sE(s) = s \cdot \frac{1}{s^2} \cdot \frac{s+a}{s+a+k_p} = \frac{a}{s(a+k_p)} = \infty$$

$$\therefore \text{Error grows unbounded}$$

Question: Why did we use FVT of error E and not output Y?

Ans: If we had been able to track a ramp, the FVT of Y would also be ∞ .

More generally:

Redefining $S = \frac{1}{1+GH}$, if S is stable, then (by FVT):

$$\text{Step: } r(t) = 1 \rightarrow e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} S(s)$$

$$\text{Ramp: } r(t) = t \rightarrow e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s} S(s)$$

$$\text{Parabolic: } r(t) = \frac{t^2}{2} \rightarrow e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s^2} S(s)$$

$$\vdots \qquad \qquad \qquad \vdots$$

$$\text{General: } r(t) = \frac{t^k}{k!} \rightarrow e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s^k} S(s)$$

→ Higher degree reference polynomials correspond to faster moving references (step → ramp → parabola etc), each modeled as an integral $\frac{1}{s}$ of the former. To track these, we'll have to design our controller or rely on our plant to cancel the $\frac{1}{s}$ terms.

∴ A stable system is of Type k if it can track polynomial signals of order k with bounded error.

Examples

Let's see the effect on steady-state error from adding integral terms to our controller.

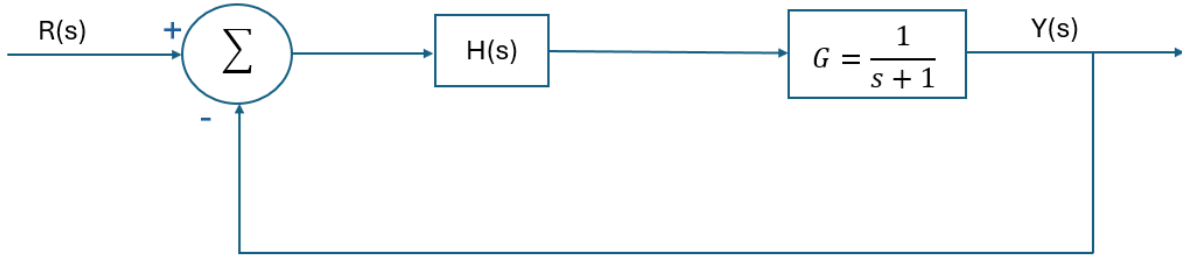


Figure 7.2: Example block diagram.

1. $H(s) = k_p$

$$S(s) = \frac{1}{1+GH} = \frac{1}{1 + \frac{k_p}{s+1}} = \frac{s+1}{s+1+k_p} \quad \} \quad \text{Stable for } k_p > -1$$

$$r(t) = 1 \quad \rightarrow \quad e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{s+1}{s+1+k_p} = \frac{1}{1+k_p} \quad \left. \vphantom{\lim_{s \rightarrow 0} \frac{s+1}{s+1+k_p}} \right\} \begin{array}{l} \text{-- Bounded non-zero error} \\ \text{-- Large } k_p \rightarrow \text{smaller error} \end{array}$$

$$r(t) = t \quad \rightarrow \quad e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s} \cdot \frac{s+1}{s+1+k_p} = \infty \quad \} \quad \text{Unbounded error (for any } k_p)$$

$\therefore$ System type = 0

2. $H(s) = \frac{k_I}{s}$

$$S(s) = \frac{1}{1+GH} = \frac{1}{1 + \frac{k_I}{s(s+1)}} = \frac{s(s+1)}{s^2+s+k_I} \quad \} \quad \text{Stable for } k_I > 0$$

$$r(t) = 1 \quad \rightarrow \quad e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} S(s) = 0 \quad \} \quad \text{Bounded error} = 0$$

$$r(t) = t \quad \rightarrow \quad e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s} \cdot \frac{s(s+1)}{s^2+s+k_I} = \frac{s+1}{s^2+s+k_I} = \frac{1}{k_I} \quad \left. \vphantom{\lim_{s \rightarrow 0} \frac{s(s+1)}{s^2+s+k_I}} \right\} \begin{array}{l} \text{-- Bounded non-zero error} \\ \text{-- Large } k_I \rightarrow \text{smaller error} \end{array}$$

$$r(t) = t^2 \quad \rightarrow \quad e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s^2} \cdot \frac{s(s+1)}{s^2+s+k_I} = \frac{s+1}{s(s^2+s+k_I)} = \infty \quad \} \quad \text{Unbounded error (for any } k_I)$$

$\therefore$ System type = 1

Summary

System type is defined by the number of $\frac{1}{s}$ in GH, and dictates the degree of the polynomial it can track.

Table of Errors:

System Type	Step (position)	Ramp (velocity)	Parabola (acceleration)
0	$\frac{1}{1+k}$	∞	∞
1	0	$\frac{1}{k}$	∞
2	0	0	$\frac{1}{k}$

7.2 Routh-Hurwitz Stability Criterion

When we find our transfer functions, the first thing we want to know is if they are stable. Consider our closed loop transfer function:

$$G_{RY} = \frac{GH}{1+GH}$$

$$\left. \begin{array}{l} \text{For 1st order plant: } G = \frac{1}{s+1} \\ \text{and PI controller: } H = k_p + \frac{k_I}{s} \end{array} \right\} \text{ We have: } G_{RY} = \frac{k_p s + k_I}{s^2 + (k_p + 1)s + k_I}$$

→ Our transfer function is an (unfactored) polynomial.

Can we determine the stability of a transfer function without factoring it?

$$\text{1st Order: } \frac{1}{s+a_1} \rightarrow \text{stable if } a_1 > 0$$

$$\text{2nd Order: } \frac{1}{s^2 + a_1 s + a_2} \rightarrow \text{stable if } a_1, a_2 > 0$$

Roots, or real part of roots, are guaranteed to be in LHP.

- If just one coefficient $a_i < 0$, the system is unstable.
- *Note, this assumes we've put the polynomial in a form where highest order coefficient = +1.*
- If every term has a negative coefficient, it's stable with negative gain.

$$\rightarrow e.g. \frac{1}{-s^2 - a_1 s - a_2} = \frac{1}{-(s^2 + a_1 s + a_2)} = \frac{-1}{s^2 + a_1 s + a_2}$$

What about higher order systems (hard to factor manually)? [E.g. when we do PI control of a 2nd order plant we get a 3rd order polynomial.]

→ Routh Hurwitz Criterion

- Let's us assess stability without solving for roots of s.
- Guarantees all roots in LHP if certain combinations of coefficients satisfy a condition.

For a polynomial $s^k + a_1 s^{k-1} + a_2 s^{k-2} + \dots$

if any $a_i < 0$ in the polynomial, we know the system is unstable [because this can only occur for a negative in root $(\frac{1}{s-a})$ i.e. RHP].

What about when all coefficients are positive?

$$\begin{array}{ccc} \overbrace{\frac{1}{(s+b_1)(s+b_2)(s+b_3)(s+b_i)\dots}}^{\text{Our poles}} & \begin{array}{c} \checkmark \\ \rightarrow \\ \leftarrow \\ \times \end{array} & \overbrace{\frac{1}{s^k + a_1 s^{k-1} + a_2 s^{k-2} + \dots}}^{\text{Our polynomial coefficients}} \end{array}$$

✓ if all $b_i > 0$ (LHP) → all $a_i > 0$ ∴ stable

× if all $a_i > 0$ → no guarantee all $b_i > 0$ (LHP) because of complex roots in RHP

$$\begin{array}{ccc} \text{E.g: } \frac{1}{s^4 + 2s^3 + 3s^2 + 10s + 8} & \rightarrow & \underbrace{\left(\frac{1}{s^2 - s + 4}\right)}_{0.5 \pm j\sqrt{3.75}} \underbrace{\left(\frac{1}{s+2}\right)\left(\frac{1}{s+1}\right)}_{\text{both in LHP}} \\ & & \rightarrow 2 \text{ in RHP} \end{array}$$

- Routh criteria for 3rd-order polynomials

$$s^3 + a_1 s^2 + a_2 s + a_3 \rightarrow \text{LHP roots if: } \begin{cases} a_1 > 0, a_2 > 0, a_3 > 0 \\ a_1 \cdot a_2 > a_3 \end{cases}$$

- Criteria for higher order polynomials can be looked up or computed using the "Routh array".